

ENGINEERS WITHOUT BORDERS-USA, UNIVERSITY OF KANSAS CHAPTER

DRAFT HYDROGEOLOGICAL SURVEY REPORT FOR

SITING OF 01 BOREHOLE SITE IN BUBAGO VILLAGE, BUWOOYA SUBCOUNTY, MAYUGE DISTRICT

MARCH 2023

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ABBREVIATIONS AND GLOSSARY OF TERMS

ABBREVIATIONS (S.I. Units throughout, unless indicated otherwise)

agl	above ground level
NWSC	National water and Sewerage corporation
amsl	above mean sea level
bgl	below ground level
d	day
ND	Internal diameter
OD	Outer diameter
E	East
EC	electrical conductivity(mS/cm)
H	head
hr	hour
K	hydraulic conductivity(m/day)
l	litre
m	meter
mm	millimetre
N	North
PVC	Poly vinyl chloride
DHRH	Developing Human Resource for Health
PAWL	pumped water Level
Q	discharge (m^3/hr)
Q/s	specific capacity (discharge-draw down ratio; in m^3
s	draw down(m)
S	south
sec	second
SWL	static water level
T	Transmissivity,T (m^2/day)
VES	Vertical Electrical Sounding

W	West
WSS	water struck level
MS/cm	micro-Siemen per centimeter: Unit for electrical conductivity
°C	degrees Celsius: Unit for temperature
Ωm:	Unit for Apparent resistivity
“	Inch

GLOSSARY OF TERMS

Alluvium General Term for detrital material deposited by flowing water.

Aquifer A Geological formation or structure, which stores and transmits water to wells, boreholes or springs.

Colluvium General term for detrital material deposited by hill slope gravitational processes with or without water as an agent. Usually of mixed Texture.

Conductivity Transimissivity, T per unit length (m/day).

Confined aquifer A formation in which the groundwater is isolated from the atmosphere by impermeable geologic formations. Confined water is generally at greater pressure than atmospheric, and will therefore rise above the struck level in a borehole.

Development In borehole engineering, this is the general term for procedures applied to repair the damage done to the formation during drilling. Often the borehole walls are partially clogged by an impermeable “wall cake”, consisting of fine debris crushed during drilling, and clays from the penetrated formations. Well development removes these clayey cakes, and increases the porosity and permeability of the materials around the intake portion of the well as a result, a higher sustainable yield can be achieved.

Evaporation Loss of water from a land area through transpiration from plants and evaporation from the surface.

Fault A larger fracture surface along which appreciable displacement has taken place.

Gneiss Irregularly banded rock, with predominant quartz, feldspar and micaceous minerals. A product of regional metamorphism, especially of the higher grade.

Gradient The rate of change in total head per unit of distance, which causes flow in the direction of the lowest head.

Granulate Even-grained metamorphic rock of coarse grain composed mainly of hornblende and plagioclase minerals. A product of regional metamorphism of the highest grade.

Heterogeneous Not uniform in structure or composition

Hydraulic head Energy contained in a water mass-produced by elevation, pressure or velocity.

Hydrogeological Those factors that deal with subsurface waters and related geological aspects of surface waters.

Infiltration Process of water entering the soil through the ground surface.

Joint Fractures along which no significant displacement has taken place.

Metamorphic Referring to rocks that have been changed in structure and mineralogy by a combination of high pressure, high temperature, and/or chemical factors often linked to major regional tectonic events. Examples are gneisses, schist's, granulites, etc.

Percolation Process of water seeping through the unsaturated zone, generally from a surface source to the saturated zone.

Perched aquifer Unconfined groundwater separated from an underlying main aquifer by an unsaturated zone. Downward percolation hindered by an impermeable layer.

Permeability The capacity of a porous medium to transmit fluids.

Piezometric level An imaginary water table, representing the total head in a confined aquifer, and is defined by the level to which water would rise in a well.

Porosity The portion of bulk volume in a rock or sediment that is occupied by openings, either isolated or connected.

Pumping test A test that is conducted to determine aquifer and/or well characteristics.

Recharge General term applied to the passage of water from surface or subsurface sources (e.g. rivers, rainfall, lateral groundwater flow) to the aquifer zones.

Regolith The mantle of loose or semi-consolidated material (soils, sediments and broken rock) overlying the bedrock.

Saprolite Weathered residual rock in place.

Static water level the level of water in a well that is not being affected by pumping. (Also known as "rest water level").

Transmissivity, T A measure for the capacity of an aquifer to conduct water through its saturated thickness (m^2/day).

Unconfined Referring to an aquifer situation whereby the water table is exposed to the atmosphere through openings in the overlying materials (as opposed to confined conditions).

Yield Volume of water discharged from a well in a unit time.

CONTENTS

1.0	PREAMBLE	<u>6</u>
2.0	INTRODUCTION	6
2.1	Objectives and General project Approach	7
2.1.1	Objective.....	7
2.2	Topography, Climate and Vegetation	8
2.3.	Drainage	9
3.0	GENERAL GEOLOGY	9
4.0	CONCEPTUAL HYDROGEOLOGY	10
5.0	PRINCIPLE OF ELECTRICAL RESISTIVITY METHOD	11
6.0	METHODOLOGY OF INVESTIGATION	12
6.1	Desk study	12
6.2	Resistivity Profiling.	12
6.3	Vertical Electrical Soundings (VES)	13
7.0	GEOPHYSICAL SURVEY AND INTERPRETATION	13
7.1	Results and Discussion	13
7.2	Calibration Surveys	14
8.0	CONCLUSIONS AND RECOMMENDATIONS	15
8.1	Recommendations	15
APPENDIX II: GEOPHYSICAL DATA		17
APPENDIX III: BOREHOLE DESIGN		22

List of Figures:

<i>Fig.1: Map of Uganda showing the Location of Mayuge District.</i>	7
<i>Fig.2: Digital Elevation Model (DEM) of the project area</i>	8
<i>Fig.3: Drainage map of the project area</i>	9
<i>Fig.4: Geology map of the project area</i>	10
<i>Fig.5: Schlumberger electrode array.</i>	11

List of Tables:

<i>Table 1: Summary of Recommended sites.....</i>	15
<i>Table 1: Summary of all VES data.....</i>	16

1.0 PREAMBLE

Engineers Without Border–USA University of Kansas intends to drill a borehole to supply water to Bubago community in Buwooya Sub County, Mayuge District. It's upon this background that **Rica Consult (U) Ltd** was contracted to carry out a comprehensive geophysical survey for one (01) borehole site in Bubago for the drilling of the borehole with sufficient yield.

In the current state, Bubago village is not supplied by National Water and Sewerage Corporation nor any other safe water programme. The community depends on the only existing streams within the village as well private owned hand dug wells. This puts the lives of the people at stake as the sources above are prone to contamination hence water borne diseases.

Various hand dug wells have been constructed by individuals around homesteads but have totally dried up after a very short period. Therefore an urgent need for safe portable water is needed to save the community.

Field Reconnaissance Survey was carried out concurrently with Hydrogeological and Geophysical Survey from **13th to 15th March, 2023**. Following below is a report containing details on data generated, data interpretation, recommendations and conclusions.

2.0 INTRODUCTION

Bubago village is located in Buwaiswa Sub County, Mayuge district. The district is located in the eastern part of Uganda between grids 0.626°N to -1°S and 33.27°E to 33.72°E. Mayuge district shares borders with districts of Bugweri, Iganga and Luuka to the North, Bugiri to the north east, Namayingo to the east, Republic of Tanzania to the South, Buvuma to the west as well as Jinja and Buikwe to the West. It is about 120 Km East of Kampala and the road from Kampala is Tarmac. The rest of the district has a fairly good all weather - road network.

Bubago village altitude ranges from 1125m in the Valley bottoms to about 1180m on highest peaks.

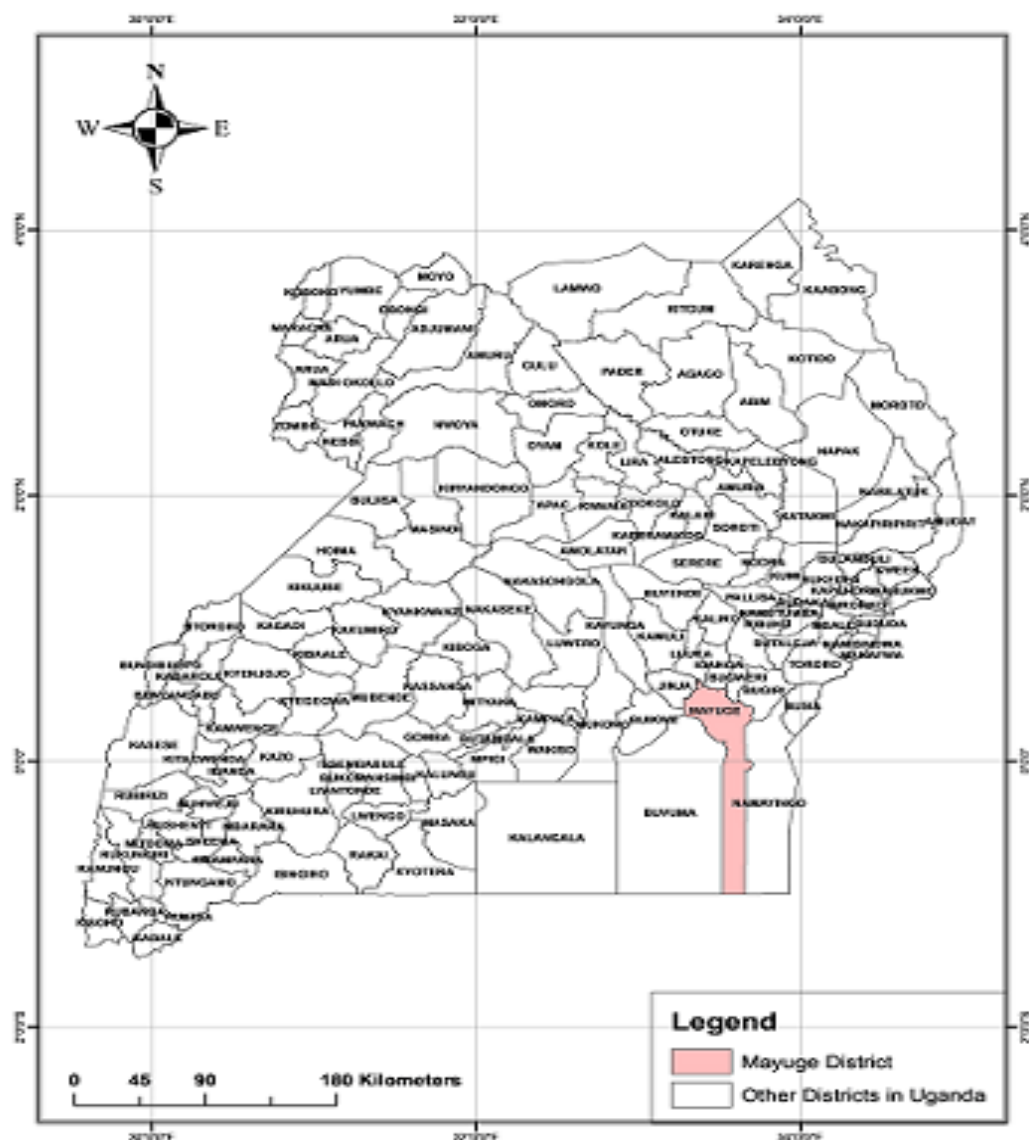


Fig.1: Map of Uganda showing the Location of Mayuge District.

2.1 Objectives and General project Approach

2.1.1 Objectives

The goal of the siting study is to provide an overview of the borehole potential of the area of interest by:

- 1) Creating a fair and extensive overview of the hydro-geological situation of the area around Bubago village and;
- 2) Ranking all potential drilling sites within this area of interest.

It's against this background that Engineers without Borders-USA University of Kansas Chapter hired the Services of Rica Consult (U) Ltd to carry out

Hydrogeological and Geophysical survey for purposes of establishing a site to be drilled to form a new safe portable water source for Bubago community.

The Consultant's services generally cover the following:

- Siting for borehole, including conducting desk studies reconnaissance survey, Hydrogeological and Geophysical investigations.

2.2 Topography, Climate and Vegetation

Bubago is situated in a generally hilly area, with an elevation range from 1125 to 1180 m.a.s.l. The Climate is generally warm, muggy and overcast. Over the year, the temperature typically varies from 62°F to 86°F and is rarely below 60°F or above 93°F. The vegetation type is of cropland, herbaceous vegetation and forest/savannah mosaics. The vegetation of the area has suffered from pressure created by rapid population growth and demand for fuel, farmland, settlement and timber. The soil types comprising the study area are mostly sandy clay loams, black cotton and sandy alluvial soils.

The annual rainfall of the study area falls in the range of 1200 mm.

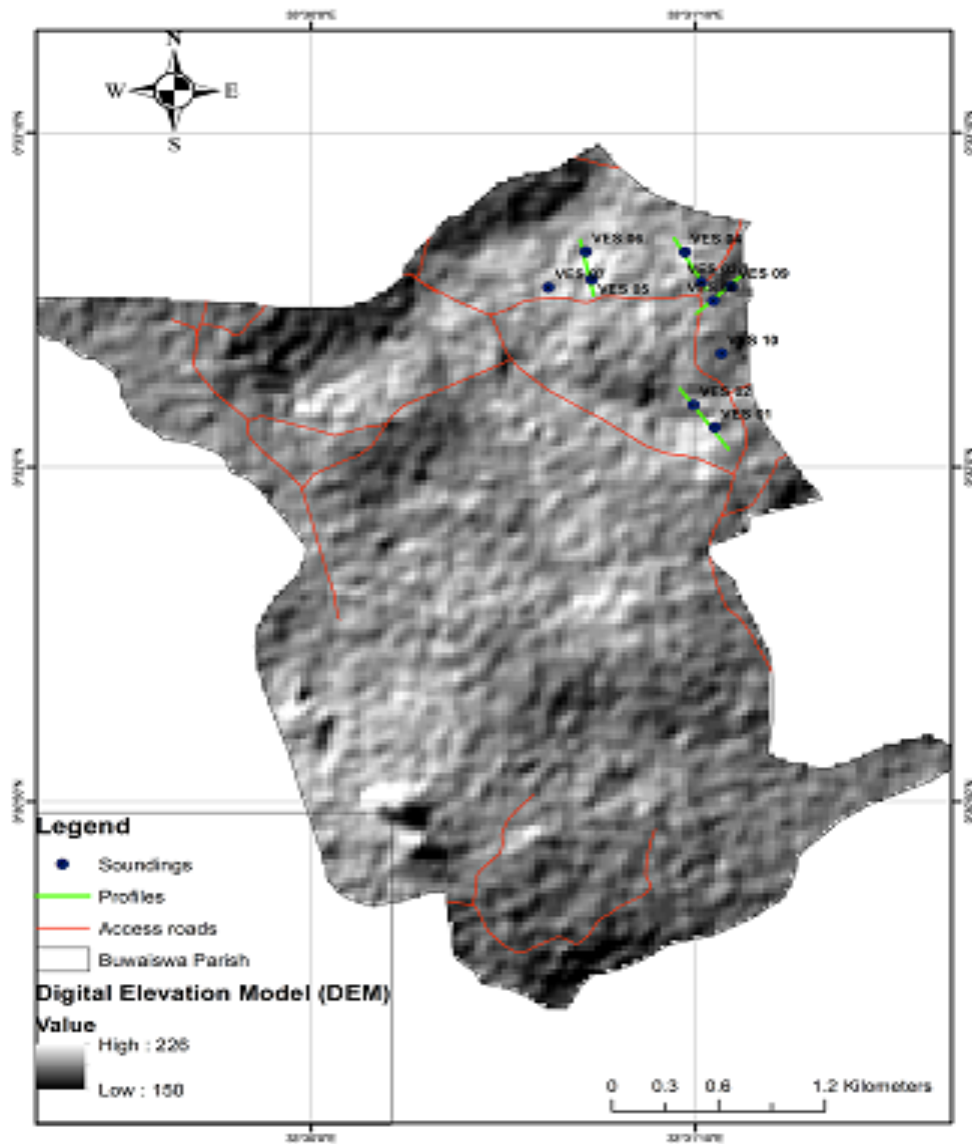


Fig.2: Digital Elevation Model (DEM) of Bubago project area

2.3. Drainage

In general, the drainage of Bubago area is not very defined. It is comprised of two small streams draining the area one in the eastern part of the village draining southwards and the other in the northern part draining westwards. No major wetland in the project area.

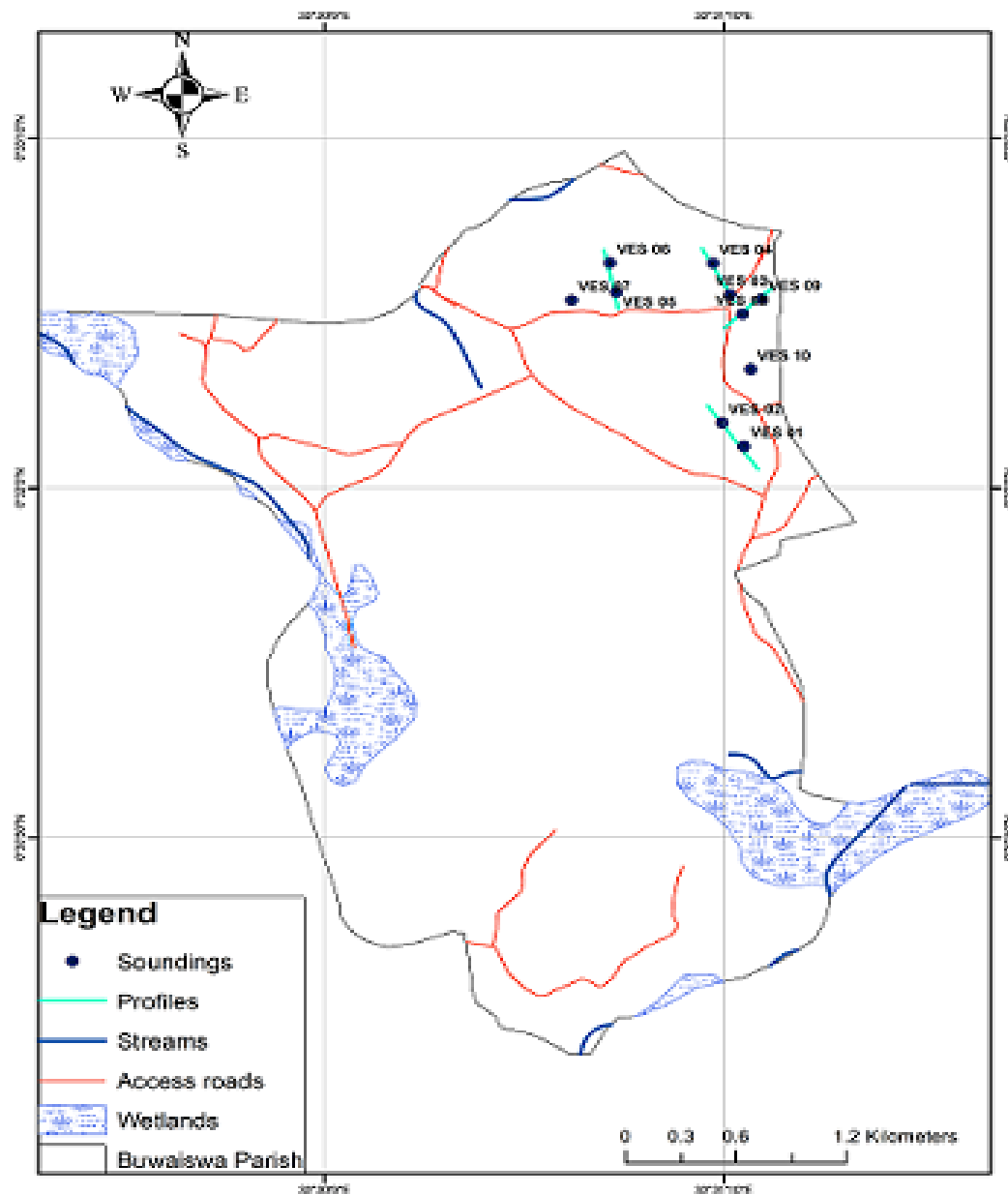


Fig.3: Drainage map of Bubago project area

3.0 GENERAL GEOLOGY

According to the Geological Map of Uganda (Compiled by R. MacDonald and drawn by Department of Geological Surveys and Mines; 1966 scale: 1:1,500,000 as supported by the available shape files), the entire project area is predominantly underlain by Basement complex comprised of medium grained granite rock. The Geological Map of the project area is depicted in figure 4 below.

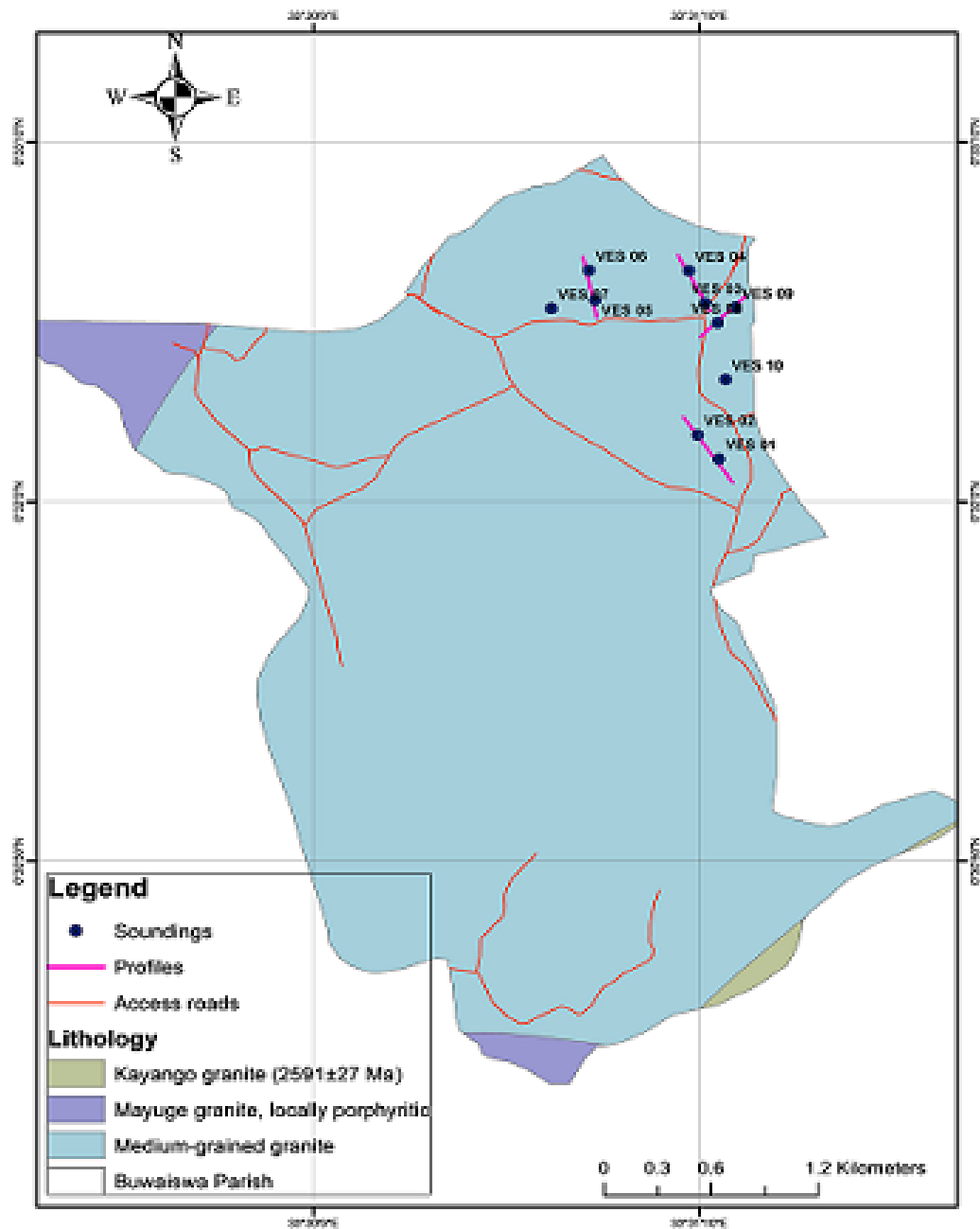


Fig.4: Geology map of Bubago project area

4.0 CONCEPTUAL HYDROGEOLOGY

Groundwater generally occurs in the weathered rocks or overburden (regolith) and in the fractured rock. The weathered rock may have a good transmissivity and storage abilities to provide some yield; however, generally the better aquifers are found in the contact between the overburden and the fresh rock. This zone is mechanically disintegrated with less secondary clay minerals resulting in a higher transmissivity. Ultimately, the highest yielding aquifers can be expected in the

fracture bedrock. Boreholes are usually drilled into the fractured bedrock where the permeability is rather high and where the storage can be provided by the overburden. Large and deep aquifers can be recharged through a connected system of fractured zones. The recharge of shallow aquifers, found in the overburden or in a fractured upper part of the bedrock is generally dependent on the size of the catchment area and the lithological character of the bedrock.

From the existing shallow hand dug wells in Bubago project area it indicates saturated thin regolith and the transition zone as the main productive zones. The general lithology in the area is brown top soil, clay and granite rock.

5.0 PRINCIPLE OF ELECTRICAL RESISTIVITY METHOD

A direct current is sent through the ground between two metal electrodes. Since different earth layers offer resistance to the passage of current in varying degrees, some voltage loss will occur as the current flows from one electrode to the other. The Voltage loss is measured at potential electrodes placed between current electrodes. Conductivity of current in the subsurface depends on the conductivity of the formation encountered, porosity, as well as the volume of water in the pores. The presence of water and its chemical character are the principal controls on the flow of electric current because most rock particles offer high resistance to current flow. This is done by introducing low frequency direct current into the ground through two steel electrodes, designated A and B.

The resulting potential distribution on the ground, which is a measure of the electrical resistivity of the sub-surface materials, will then be measured by means of two other steel probes or electrodes, M and N, as shown below:

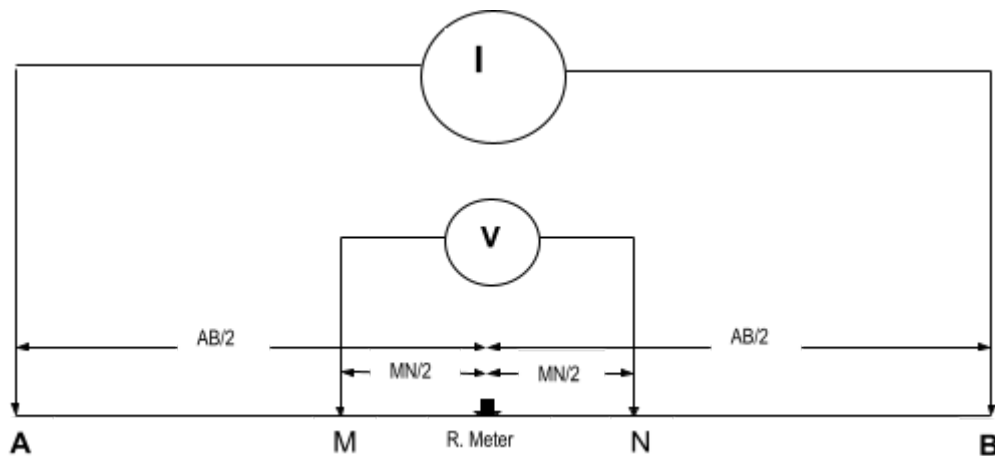


Fig5: Schlumberger electrode array.

Current and potential electrode positions are marked from the measuring point of the resistivity meter such that $AB/2$ and $MN/2$. Where $AB/2$ = Current electrode spacing and $MN/2$ = Potential electrode spacing.

The electrodes are driven into the earth in a straight line to make a good contact with the earth. The current electrodes spacing are expanded over a range of values for measurements in the field. The values of $AB/2$ increase as the measurements progress while the potential electrodes separations are guided accordingly. The potential electrodes are kept at small separations relative to the current electrode separations. During the field work taking a reading, the resistivity meter performs automatic recording of both voltage (V) and current (I), stacks the results, computes the resistance in real time and digitally displays it. The apparent resistivity value is the product of the geometric factor and the resistance recorded in the resistivity meter.

Apparent values of resistivity plotted against current electrode spacing ($AB/2$) gives a characteristic curve that is interpreted.

Resistivity decreases with increasing saturation and salinity of the water in the pore. In all the locations in the project area, resistivity values were in normal ranges with no likelihood of saline water.

6.0 METHODOLOGY OF INVESTIGATION

6.1 Desk study

The work involved a preliminary desktop study. The study mainly involved generating/studying GIS maps covering the area (from the available shape files) and analyzing scanty available data on the existing water sources in the project area. This analysis provided information on water quality, ground formations, water bearing zones and depth to water, yield of the sources and hydrology of the area for purposes of assessing ground water recharge.

It was concluded from the desk study that the most favorable areas for ground water exploration in the project area are the SW-NE and N-S trending valleys that form the village boundaries. These are structurally/topographically favored as well as possess the most pronounced lineament patterns. This was targeted for the geophysical surveys (figure 2).

Investigations involved using a GPS (Way point) to identify the actual orientation on the ground of the preferred lineaments as seen from the Digital Elevation model as well as running profiles (Electric and Dowsing) to identify areas (points/locations) of thick water saturated formations (Anomalies).

6.2 Resistivity Profiling.

The aim of resistivity profiling is to map lateral anomalous resistivity zones. These may be caused by the presence of a fractured zones and / or a deeper weathered section that may be favorable for groundwater abstraction. Anomalous measurements were expected near the main topographical lineament (valley). Profiles were carried out using a station interval of 10 m and $1/2AB$ of 83 m. The

profiling was not done across the valley structure but rather utilized the community preferred areas.

6.3 Vertical Electrical Soundings (VES)

Vertical electrical soundings were subsequently carried out on the anomalous (low values) stations on the profiles. The soundings were carried out to get an insight in the differences in depth and type of the overburden. It should be stressed that in the absence of well log data of boreholes drilled on surrounding areas, the interpretations of depths of layers are to be considered more quantitatively than qualitatively. For example, the interpreted depth to the fresh bedrock becomes larger if a higher resistivity for the pre-bedrock layer is used for the interpretation model.

The results of the survey are given in the siting sheets compiled in the Appendix I.

7.0 GEOPHYSICAL SURVEY AND INTERPRETATION

On the basis of the methodology and theoretical principles of the methodology, both of which are already given above, the strategy was to identify sites with multi-layered aquifer systems using Geophysics and Hydrogeological methods in order to select the best sites for the deep boreholes. Sites were surveyed using the Topographic Map, GPS, Water Dowsing and Geo-electrical Resistivity Sounding methods. Major ground water flow directions were carefully studied and taken into consideration before making the final site selection.

It should be noted that interpretations are mere simplifications of the heterogeneity of the sub-surface, which consequently means that some of the interpretations may be differing from the well logs generated during drilling.

7.1 Results and Discussion

The interpretation of the sounding curves was done both qualitatively and quantitatively. The qualitative interpretation entails the observation of the sounding curves as plotted on the bi-logarithm graph paper. Four short profiles of **1.52 Km** long and ten (**10**) **VESs** were conducted in the study area using the schlumberger array. The geo-electric profiles carried out on the higher altitude areas of did not result in pronounced anomalies (more resistive). Only profiles on the lower sections of the village resulted in low resistive anomalies. The results revealed the presence of multi-layered lithological formations in the surveyed area.

The survey data was interpreted and conclusions made basing on “Ressound simulation model interpretation”, “Geoform” data presentation and expert knowledge.

Table 1 below summarizes details of the survey interpretation. The site recommended for drilling is **VES 06**. It is clearly marked with a wooden peg for easy identification. Alternative site to the recommended site (**VES 04**) too is clearly marked. It should be noted that an alternative site shall be considered for drilling only in case of a technical failure at the recommended site.

Table 1: Summary of recommended sites

VES ID/Details	GPS Grids	DTW (m)	DTBR (m)	RDD (m)	Expected Formation
VES-06 (RECOMMENDED SITE)	E: 557183 N: 60335 Alt: 1130 m	19-40	19	95	Top soil, Sandy clay, Weathered rock, Fractured bedrock
VES-04 (ALTERNATIVE SITE)	E: 557742 N: 60333 Alt: 1143 m	27-60	40	100	Top soil, Sandy clay, Weathered rock, Fractured bedrock

It should be noted that the project area is relatively shallow bed rock as well as thin weathered zones hence saturated. The abnormally high apparent resistivity values on the lower station intervals on the profiles are attributed to the granite ridges where most profiling started. A comparatively low resistivity anomaly will most probably correspond to fresh water and therefore less resistive sections of the bedrock were targeted in this case for groundwater abstraction. It is on this basis mainly that the sites for sounding were selected from the profiles. Other factors considered include; expert knowledge of the area, land ownership, accessibility, etc.

7.2 Calibration Surveys

In addition to interpretations, the apparent resistivity measurements carried out at an existing borehole (both yielding and dry) assist in analysis of the collected data. Calibration is done for comparison purposes only. No calibration VESs were carried out due to lack of existing boreholes in the project area.



Recommended site visit

8.0 CONCLUSIONS AND RECOMMENDATIONS

The objective of this survey was to investigate whether there was any good groundwater potential point within the project area for drilling a new borehole to provide safe portable water for domestic use.

A thorough investigation was carried out and a drilling site was identified at **VES-06**. The geophysical survey results of the final recommended site are much more promising and have thus been selected for drilling.

8.1 Recommendations

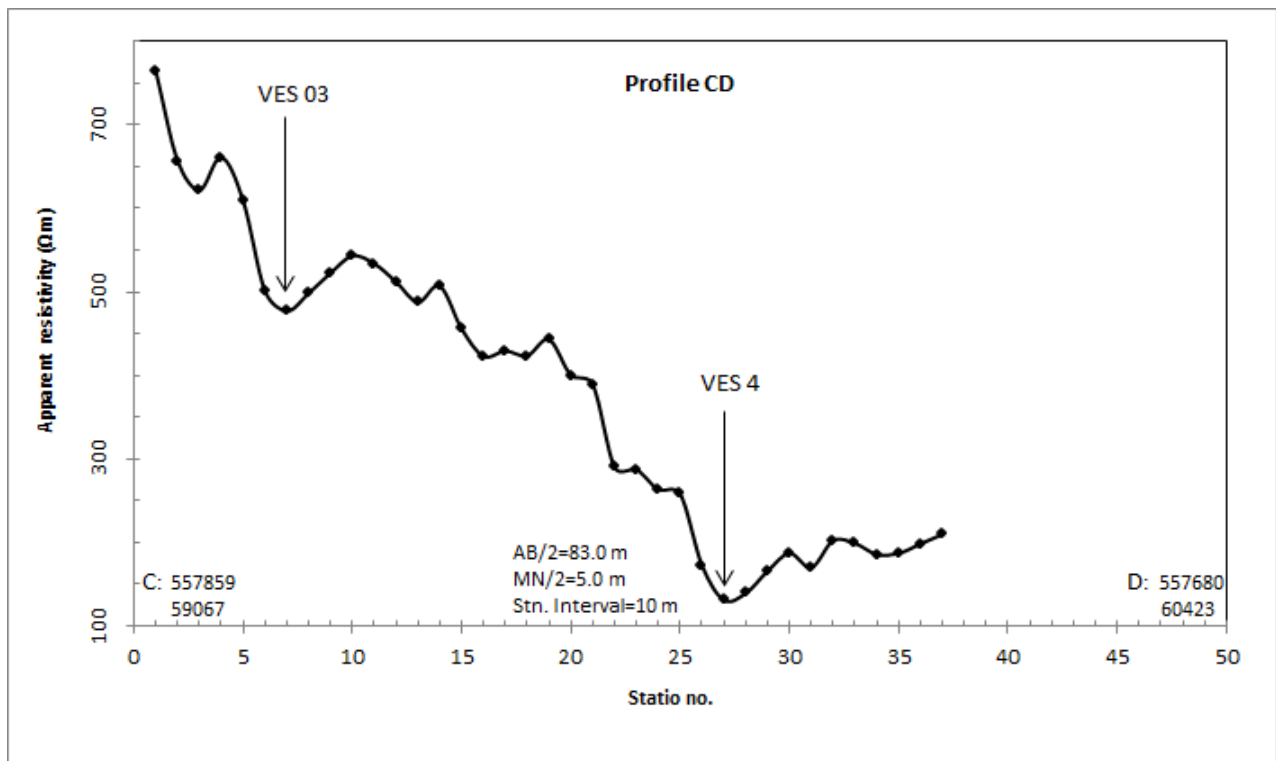
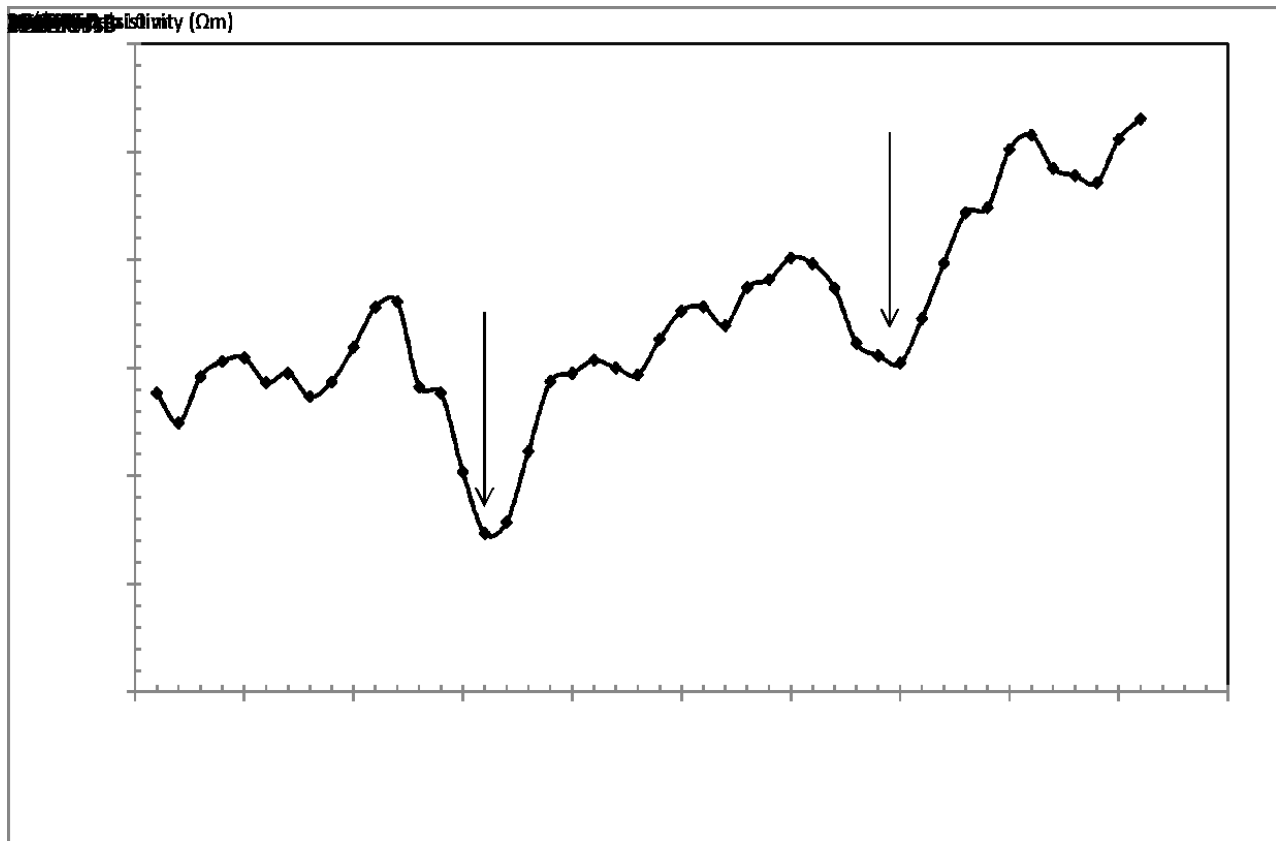
Drilling should be done at the exact recommended site and to the recommended depth unless the drilling supervisor considers deviating from the recommendations based on the encountered drilling challenges.

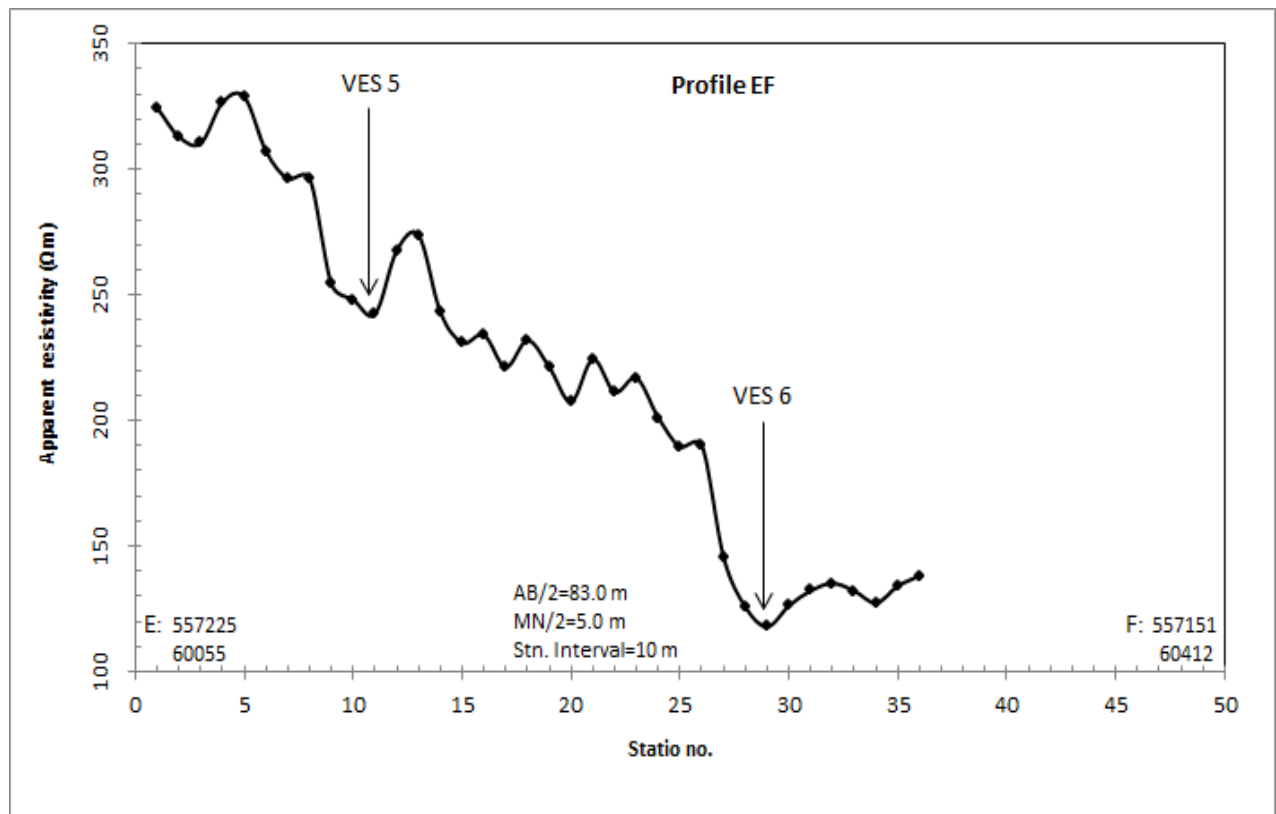
The drilling exercise should be implemented as soon as possible before the rainy season escalates. The accessibility of the above sites may be compromised by the rains.

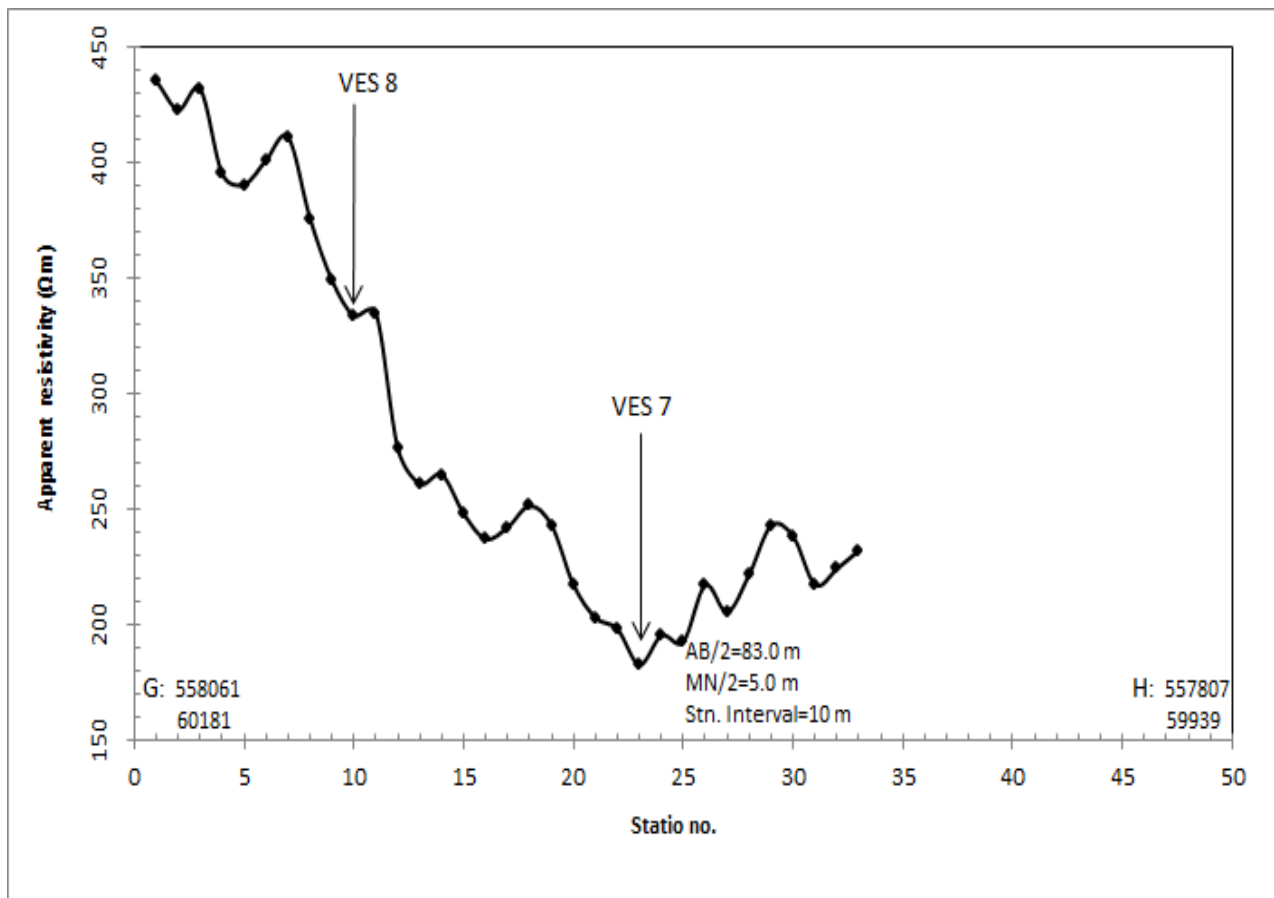
PVC Casings for all the drilled length should be installed in a way such that a maximum of 2 screens are joined together. The Construction Supervision of such borehole requires a highly experienced **Hydrogeologist**. Borehole drilling shall effectively be **air-Rotary** drilling method. An experienced driller should be engaged in this highly complex formation. The recommended borehole design (**Shallow well design**) is attached in Appendix III. The final design applied will depend on underground conditions encountered during borehole construction.

- Borehole evaluation (test pumping) will be carried out after carrying out to determine the borehole efficiency and hydraulic parameters like Storativity, Hydraulic Conductivity and Optimum Safe Pumping rate for long periods of water abstraction.
- Water quality analysis should be done on water from the well to ascertain its suitability for human consumption.
- The submersible pump to be installed in future will depend on the established safe pumping rate after borehole evaluation, and, head to be overcome. This has to be done after careful interpretation of the well evaluation results to ensure maximum borehole operation efficiency.

APPENDIX I: GEOPHYSICAL SURVEY DATA (PROFILES AND SOUNDINGS)



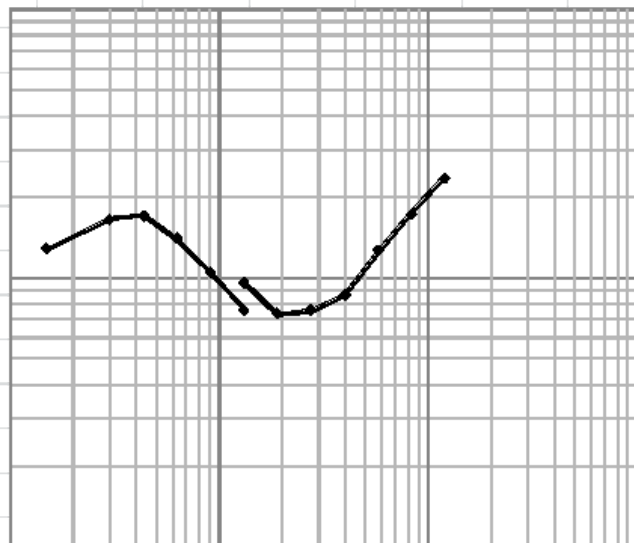




GEOPHYSICAL SURVEY DATA

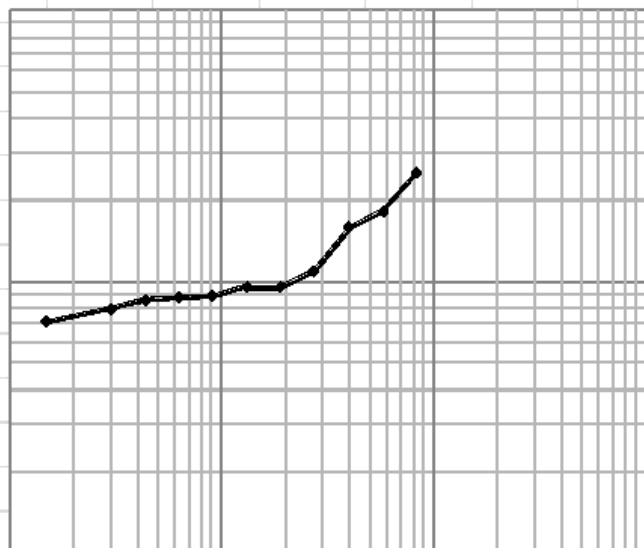
Village	Bubago				Grid E	557910	
Parish	Buwai swa				Grid N	59207	
Subcounty	Buwoya				Alt (m)	1169	
District	Mayuge						
Date	13-Mar-23						

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	f_{app} (Ω m)
1.50	0.50	20.370	6.28	127.92
3.00	0.50	6.025	27.48	165.54
4.40	0.50	2.840	60.01	170.41
6.30	0.50	1.130	123.84	139.94
9.10	0.50	0.405	259.24	104.99
13.20	0.50	0.139	546.33	75.94
13.20	5.00	2.034	46.86	95.32
19.00	5.00	0.693	106.00	73.46
27.50	5.00	0.329	230.00	75.67
40.00	5.00	0.174	494.55	86.05
58.00	5.00	0.120	1049.00	125.88
83.00	5.00	0.080	2156.00	172.48
120.00	5.00	0.052	4516.00	234.83



Village	Bubago				Grid E		557791	
Parish	Buwai swa				Grid N		59252	
Subcounty	Buwoya				Alt (m)		1159	
District	Mayuge							
Date	13-Mar-23							

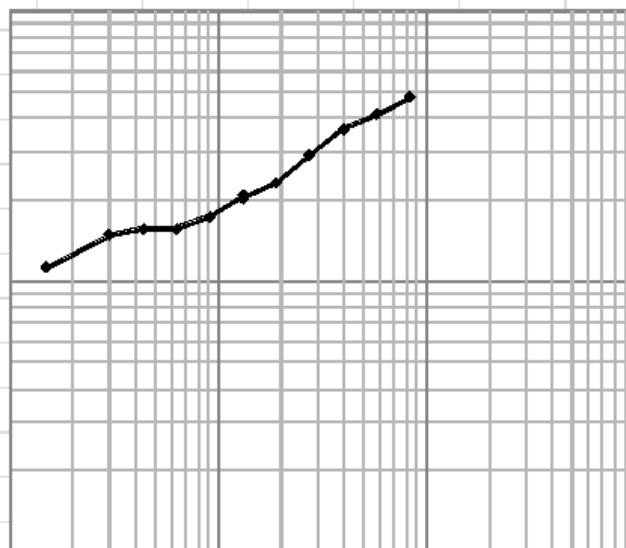
AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	J_{app} (Ωm)
1.50	0.50	11.300	6.28	70.96
3.00	0.50	2.890	27.48	79.40
4.40	0.50	1.415	60.01	84.91
6.30	0.50	0.701	123.84	86.81
9.10	0.50	0.340	259.24	88.14
13.20	0.50	0.175	546.33	95.61
13.20	5.00	2.027	46.86	94.99
19.00	5.00	0.898	106.00	95.19
27.50	5.00	0.475	230.00	109.25
40.00	5.00	0.319	494.55	157.76
58.00	5.00	0.173	1049.00	181.48
83.00	5.00	0.117	2156.00	252.25
120.00	5.00	0.000	4516.00	0.00



GEOPHYSICAL SURVEY DATA

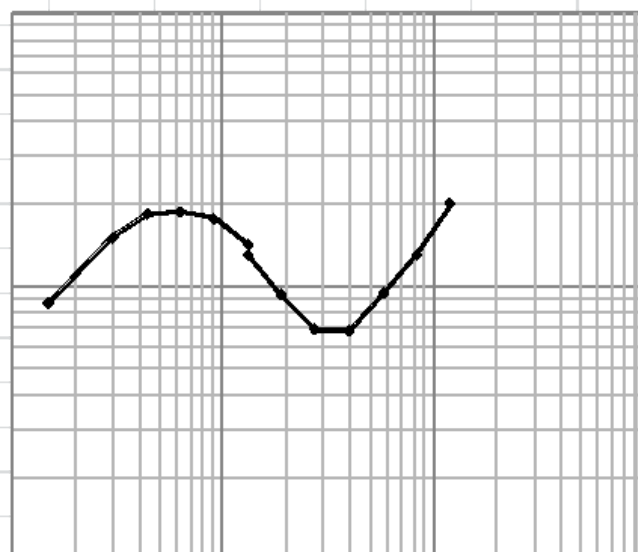
Village	Bubago				Grid E		557838	
Parish	Buwai sw a				Grid N		60138	
Subcounty	Buwoya				Alt (m)		1146	
District	Mayuge							
Date	13-Mar-23							

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	$\int_{app} (\Omega m)$
1.50	0.50	17.840	6.28	112.04
3.00	0.50	5.400	27.48	148.37
4.40	0.50	2.590	60.01	155.41
6.30	0.50	1.260	123.84	156.04
9.10	0.50	0.664	259.24	172.13
13.20	0.50	0.380	546.33	207.60
13.20	5.00	4.336	46.86	208.19
19.00	5.00	2.187	106.00	231.82
27.50	5.00	1.269	230.00	291.87
40.00	5.00	0.740	494.55	365.97
58.00	5.00	0.394	1049.00	413.31
83.00	5.00	0.222	2156.00	478.63
120.00	5.00	0.000	4516.00	0.00



Village	Bubago				Grid E		557791
Parish	Buwai swa				Grid N		59252
Subcounty	Buwoya				Alt (m)		1159
District	Mayuge						
Date	13-Mar-23						

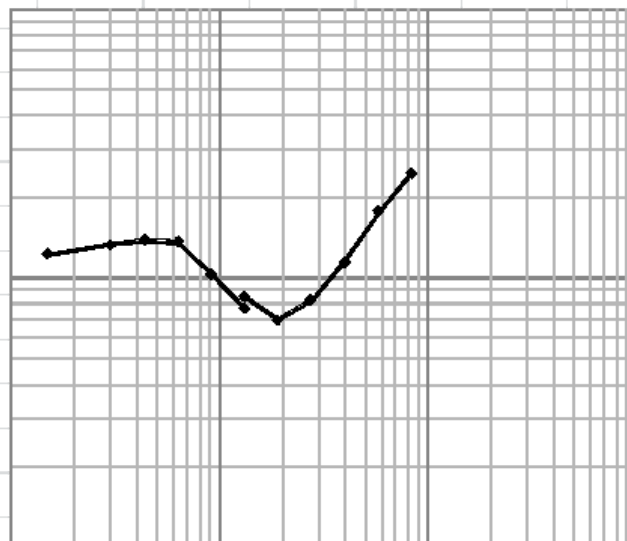
AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	J_{app} (Ωm)
1.50	0.50	13.780	6.28	86.54
3.00	0.50	5.480	27.48	150.56
4.40	0.50	3.040	60.01	182.42
6.30	0.50	1.512	123.84	187.25
9.10	0.50	0.678	259.24	175.76
13.20	0.50	0.260	546.33	142.05
13.20	5.00	2.778	46.86	130.18
19.00	5.00	0.870	106.00	92.22
27.50	5.00	0.301	230.00	69.23
40.00	5.00	0.139	494.55	68.74
58.00	5.00	0.089	1049.00	93.36
83.00	5.00	0.060	2156.00	129.36
120.00	5.00	0.044	4516.00	198.70



GEOPHYSICAL SURVEY DATA

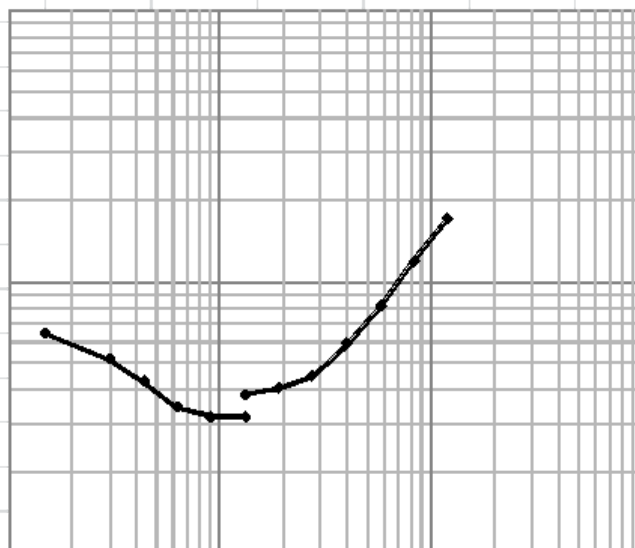
Village	Bubago				Grid E		557216	
Parish	Buwai swa				Grid N		60157	
Subcounty	Buwoya				Alt (m)		1140	
District	Mayuge							
Date	14-Mar-23							

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	$\int_{app} (\Omega m)$
1.50	0.50	19.510	6.28	122.52
3.00	0.50	4.834	27.48	132.81
4.40	0.50	2.294	60.01	137.65
6.30	0.50	1.091	123.84	135.11
9.10	0.50	0.400	259.24	108.70
13.20	0.50	0.140	546.33	76.49
13.20	5.00	1.806	46.86	84.63
19.00	5.00	0.655	106.00	69.43
27.50	5.00	0.358	230.00	82.34
40.00	5.00	0.231	494.55	114.24
58.00	5.00	0.168	1049.00	176.23
83.00	5.00	0.113	2156.00	243.63
120.00	5.00	0.000	4516.00	0.00



Village	Bubago				Grid E		557183	
Parish	Buwai swa				Grid N		60335	
Subcounty	Buwoya				Alt (m)		1130	
District	Mayuge							
Date	14-Mar-23							

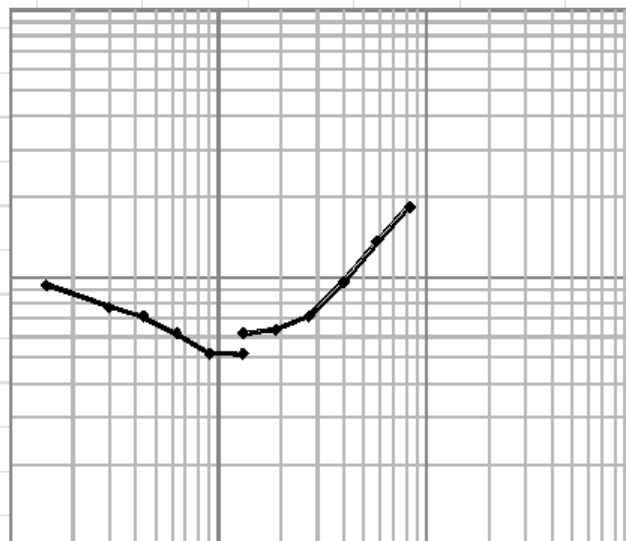
AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	J_{BPP} (Ωm)
1.50	0.50	10.220	6.28	64.18
3.00	0.50	1.876	27.48	51.54
4.40	0.50	0.714	60.01	42.84
6.30	0.50	0.277	123.84	34.30
9.10	0.50	0.122	259.24	31.63
13.20	0.50	0.058	546.33	31.69
13.20	5.00	0.818	46.86	38.33
19.00	5.00	0.380	106.00	40.28
27.50	5.00	0.194	230.00	44.62
40.00	5.00	0.119	494.55	58.85
58.00	5.00	0.077	1049.00	80.77
83.00	5.00	0.055	2156.00	118.58
120.00	5.00	0.038	4516.00	171.61



GEOPHYSICAL SURVEY DATA

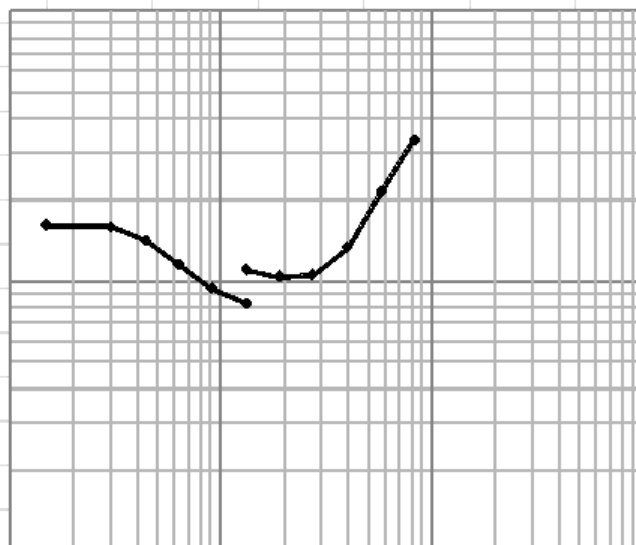
Village	Bubago				Grid E		556972	
Parish	Buwai swa				Grid N		60105	
Subcounty	Buwoya				Alt (m)		1131	
District	Mayuge							
Date	14-Mar-23							

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	f_{app} (Ωm)
1.50	0.50	14.830	6.28	98.13
3.00	0.50	2.810	27.48	77.20
4.40	0.50	1.184	60.01	71.05
6.30	0.50	0.497	123.84	61.55
9.10	0.50	0.200	259.24	51.85
13.20	0.50	0.094	546.33	51.35
13.20	5.00	1.307	46.86	61.25
19.00	5.00	0.602	106.00	63.81
27.50	5.00	0.312	230.00	71.76
40.00	5.00	0.193	494.55	95.45
58.00	5.00	0.129	1049.00	135.32
83.00	5.00	0.085	2156.00	183.26
120.00	5.00	0.000	4516.00	0.00



Village	Bubago				Grid E		557903
Parish	Buwai swa				Grid N		60019
Subcounty	Buwoya				Alt (m)		1146
District	Mayuge						
Date	14-Mar-23						

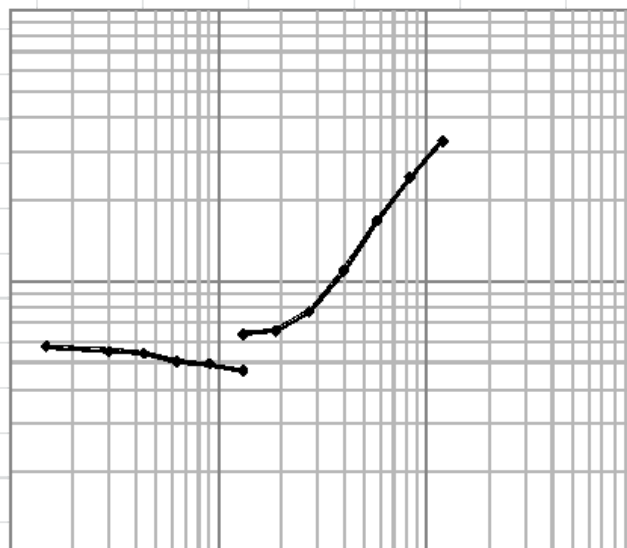
AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	$\int_{BPP}$ (Ωm)
1.50	0.50	25.600	6.28	160.77
3.00	0.50	5.808	27.48	159.57
4.40	0.50	2.361	60.01	141.67
6.30	0.50	0.934	123.84	115.67
9.10	0.50	0.362	259.24	98.84
13.20	0.50	0.152	546.33	83.04
13.20	5.00	2.347	46.86	109.98
19.00	5.00	0.978	106.00	103.67
27.50	5.00	0.458	230.00	105.34
40.00	5.00	0.268	494.55	132.54
58.00	5.00	0.205	1049.00	215.05
83.00	5.00	0.155	2156.00	334.18
120.00	5.00	0.000	4516.00	0.00



GEOPHYSICAL SURVEY DATA

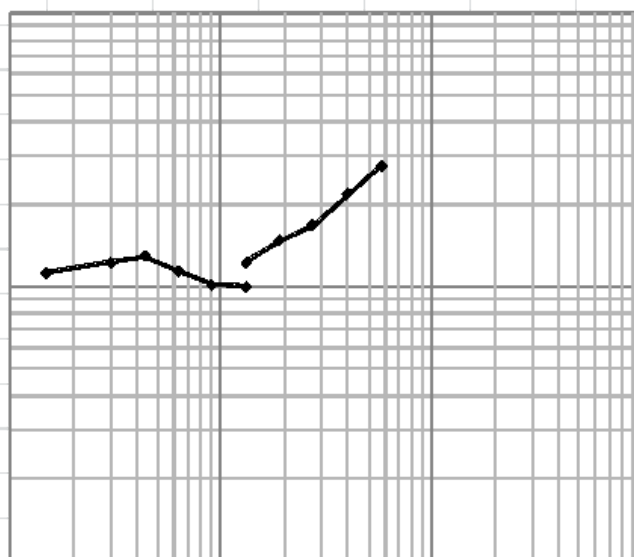
Village	Bubago	Grid E	558005
Parish	Buwaiswa	Grid N	60111
Subcounty	Buwoya	Alt (m)	1158
District	Mayuge		
Date	15-Mar-23		

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	$\int_{app} (\Omega m)$
1.50	0.50	9.080	6.28	57.02
3.00	0.50	2.000	27.48	54.95
4.40	0.50	0.903	60.01	54.18
6.30	0.50	0.406	123.84	50.28
9.10	0.50	0.190	259.24	49.26
13.20	0.50	0.085	546.33	46.44
13.20	5.00	1.360	46.86	63.73
19.00	5.00	0.619	106.00	65.61
27.50	5.00	0.336	230.00	77.28
40.00	5.00	0.220	494.55	108.80
58.00	5.00	0.160	1049.00	167.84
83.00	5.00	0.111	2156.00	239.32
120.00	5.00	0.073	4516.00	329.67



Village	Bubago	Grid E	557947
Parish	Buwaiswa	Grid N	59682
Subcounty	Buwoya	Alt (m)	1167
District	Mayuge		
Date	15-Mar-23		

AB/2 (m)	MN/2 (m)	R (Ω)	Constant K	$\int_{app} (\Omega m)$
1.50	0.50	17.860	6.28	112.16
3.00	0.50	4.460	27.48	122.54
4.40	0.50	2.150	60.01	129.01
6.30	0.50	0.920	123.84	113.98
9.10	0.50	0.390	259.24	101.10
13.20	0.50	0.183	546.33	99.98
13.20	5.00	2.620	46.86	122.78
19.00	5.00	1.390	106.00	147.34
27.50	5.00	0.730	230.00	167.90
40.00	5.00	0.440	494.55	217.60
58.00	5.00	0.264	1049.00	276.94
83.00	5.00	0.000	2156.00	0.00
120.00	5.00	0.000	4516.00	0.00



APPENDIX II: PROPOSED BOREHOLE DESIGN

